Woolyarn-seasonal-arma

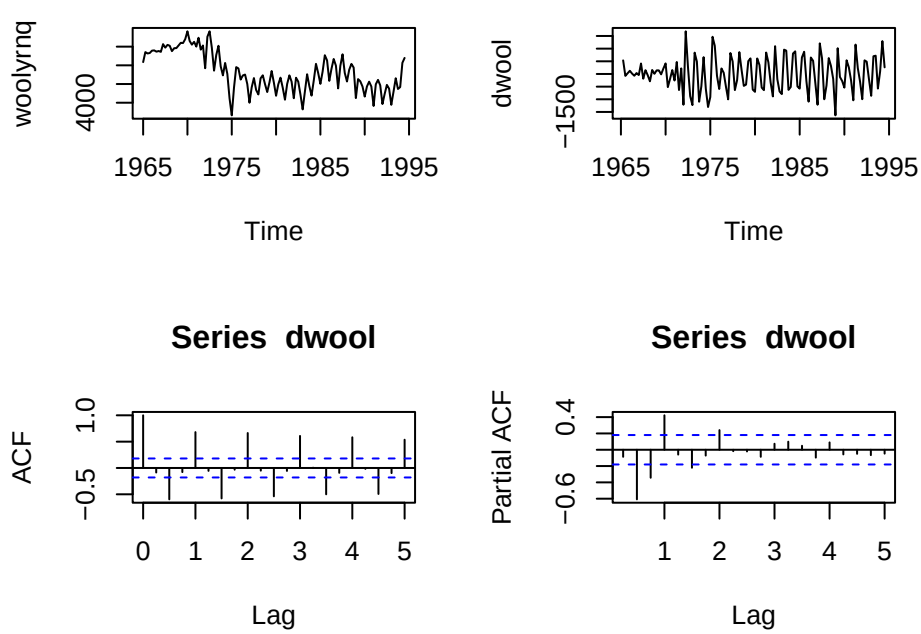
2026 March 02

Required library

```
library(forecast)
# woolyrnq: Quarterly production of woollen yarn in Australia: tonnes. Mar 1965 - Sep 1994
data(woolyrnq) # data set in library(forecast); class is "ts"
```

Set up training and holdout sets for differenced series

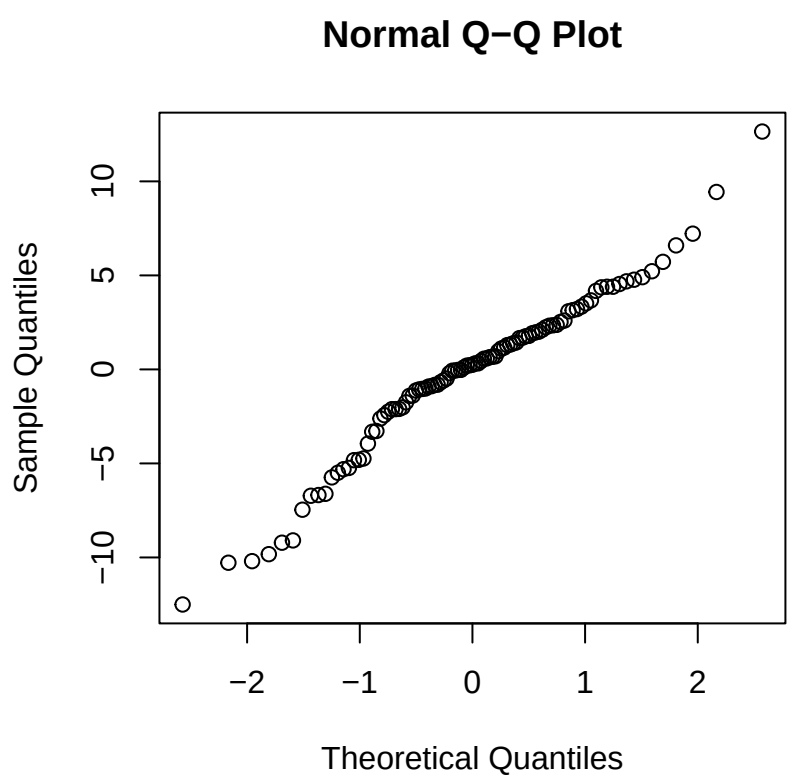
```
print(length(woolyrnq))
#> [1] 119
summary(woolyrnq)
#>    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
#>    3324    4882    5466    5658    6646    7819
par(mfrow=c(2,2))
dwool = diff(woolyrnq)
plot(woolyrnq); plot(dwool); acf(dwool)  # seasonal, lag 2 negative, lag 4 positive
pacf(dwool) #
```



```r
summary(dwool)
#>      Min.   1st Qu.    Median      Mean   3rd Qu.      Max.
#> -1635.000  -518.000    10.000     1.898   517.000  1684.000

sd(dwool)
#> [1] 680.6576

dwool_tr = dwool/100  # transform to get smaller sigma estimate

ntotal = length(dwool_tr)
# Take first 100 as training set, or first 99 after differencing.
ntrain = 99 # missing first quarter so this is 25 years, 1965-1989
train = window(dwool_tr,end=c(1989,4))
length(train)
#> [1] 99
ntrain = length(train)
holdout = window(dwool_tr,start=c(1990,1)) # 19 quarters or almost 5 years
```

Seasonal ARMA model for diff(woolyrnq)

```r
wool_auto = auto.arima(train,seasonal=T,stationary=T)

qqnorm(wool_auto$residuals)  # check normality assumption (for likelihood method)
```



```r
print(wool_auto)
#> Series: train
#> ARIMA(0,0,3)(1,0,1)[4] with zero mean
#>
#> Coefficients:
#>           ma1      ma2      ma3    sar1     sma1
#>       -0.1808  -0.1925  -0.1980  0.9454  -0.6312
#> s.e.   0.1036   0.1019   0.1207  0.0434   0.1348
#>
#> sigma^2 = 19.22:  log likelihood = -286.16
#> AIC=584.31   AICc=585.23   BIC=599.88

# Exercise: Try different seasonal ARMA models to compare AIC values.

wool_arma_03_s11 = arima(train, order=c(0,0,3),
  seasonal=list(order=c(1,0,1),period=4), method="ML", include.mean=F)
print(wool_arma_03_s11)
#>
#> Call:
#> arima(x = train, order = c(0, 0, 3), seasonal = list(order = c(1, 0, 1), period = 4),
#>     include.mean = F, method = "ML")
#>
#> Coefficients:
#>           ma1      ma2      ma3    sar1     sma1
#>       -0.1808  -0.1925  -0.1980  0.9454  -0.6312
#> s.e.   0.1036   0.1019   0.1207  0.0434   0.1348
#>
#> sigma^2 estimated as 18.25:  log likelihood = -286.16,  aic = 584.31

# Negative MA coefficients might suggest negative dependence at low lags.

# AIC is slightly smaller than AIC for previous ARMA(3,0,3) for training set.
```

The above fitted seasonal ARMA model is

$$(1-\Phi_1 B^4)Y_t = (1+\Theta_1 B^4)(1+\theta_1 B+\theta_2 B^2+\theta_3 B^3)\epsilon_t$$

as a special case of

$$\Phi(B^m)\phi(B)(Y_t-\mu) = \Theta(B^m)\theta(B)\epsilon_t,$$

when the periodicity is m.

The model can also be written as ARMA(4,7):

$$(1-0B-0B^2-0B^3-\Phi_1 B^4)Y_t = (1+\theta_1 B+\theta_2 B^2+\theta_3 B^3+\Theta_1 B^4+\theta_1\Theta_1 B^5+\theta_2\Theta_1 B^6+\theta_3\Theta_1 B^7)\epsilon_t$$

Verify h-step ahead prediction SEs

```r
pred_endoftrain = predict(wool_arma_03_s11,n.ahead=5,se.fit=T)
print(pred_endoftrain)
#> $pred
#>           Qtr1      Qtr2      Qtr3      Qtr4
#> 1990 -7.947200  9.191714  2.743480 -3.402269
#> 1991 -8.351687
```

```r
#>
#> $se
#>          Qtr1     Qtr2     Qtr3     Qtr4
#> 1990 4.272444 4.341703 4.418943 4.499172
#> 1991 4.695137

sigma2 = wool_arma_03_s11$sigma2
print(sqrt(sigma2))
#> [1] 4.272444

coe = wool_arma_03_s11$coef
print(coe)
#>        ma1        ma2        ma3       sar1       sma1
#> -0.1807875 -0.1925356 -0.1979824  0.9454051 -0.6312294

ar_coe = c(0,0,0,coe[4])
ma_coe = c(coe[1:3],coe[5],coe[5]*coe[1:3])
psiv = ARMAtoMA(ar=ar_coe, ma=ma_coe, lag.max=5)
print(psiv)
#> [1] -0.18078755 -0.19253560 -0.19798241  0.31417567 -0.05679905

psiv = c(1,psiv) # 1-step ahead SE is sqrt(sigma2)

cum_psi2 = cumsum(psiv^2)
print(cum_psi2)
#> [1] 1.000000 1.032684 1.069754 1.108951 1.207657 1.210884

# These numbers should match the abiove $se component
print(sqrt(cum_psi2*sigma2))
#> [1] 4.272444 4.341703 4.418943 4.499172 4.695137 4.701404
```